

Rina (Jingyi) Cheng

Applied Machine Learning Researcher | Demand Forecasting and Operational Decisions

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LinkedIn | GitHub | Google Scholar

Dutch permanent resident — no employer sponsorship required

PhD expected Q2 2027 | Preferred start June 2027; available from mid-March 2027

Profile

TU Delft PhD researcher developing machine learning for demand forecasting and operational decisions. Applied ML research since 2020, including collaborations with CHECK, RET and Just Eat Takeaway.com. Led modelling and product design for a live forecasting feature now available at 15 Rotterdam mobility hubs. Combines probabilistic forecasting, delivery-operations research and collaboration with industry partners.

Relevant experience

Doctoral Researcher | TU Delft

June 2023–present | Transport and Planning

Live availability forecasting for CHECK e-scooters in the RET app

- Conceived a new feature using predicted e-scooter availability to support multimodal trip planning up to 90 minutes ahead; led data acquisition, processing, modelling and initial product design.
- Developed an XGBoost forecasting service with predictions at 15-minute intervals. Coordinated with CHECK as historical-data provider, RET as app host, and Siemens Mobility for real-time API deployment.
- Helped grow the 2025 pilot into a regular RET app feature, expanded to 15 Rotterdam mobility hubs. Within three months of release, more than 75% of surveyed users rated the feature positively for usefulness.
- Designed and analysed user surveys; feedback indicated greater confidence planning journeys, particularly at night when public transport is less available.

T-STAR probabilistic demand forecasting

- Developed a contextual two-stage Transformer for sparse demand at 15-minute resolution, evaluated across 235 Capital Bikeshare stations.
- Reduced pickup-demand MAE by 19.7% and mean interval score by 50.3% versus a time-series Transformer benchmark; evaluated spatial and temporal robustness and zero-shot transfer to unseen service areas.
- Outperformed benchmarks including TFT and DeepAR. Published in Transportation Research Part C in 2026, with a public code implementation.

Multi-task demand and fleet-availability forecasting

- Developed an adaptive graph-based model jointly predicting pickups, drop-offs, fleet size and low-battery vehicle counts up to four hours ahead; evaluated point and probabilistic forecasts against temporal, spatial and multi-task benchmarks.

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Relevant experience continued

Research Assistant | TU Delft

December 2022–June 2023 | Part-time

Meal-delivery research collaboration with Just Eat Takeaway.com

- Proposed a dual-control deep reinforcement-learning framework for real-time order dispatching and idle-courier steering, using predicted demand to inform fleet decisions.
- Designed iteratively trained dispatching and steering policies to improve delivery efficiency and fairness in job assignment; shared research findings with industry collaborators.

Research Assistant | Erasmus School of Economics

August 2020–August 2022 | Research internship

- Developed short-term meal-delivery demand forecasts using quantile regression forests in an honours research collaboration with Just Eat Takeaway.com.
- Proposed demand-informed dynamic zoning to support real-time assignment and route planning; evaluated point and distributional forecasts on European and Taiwanese case studies.

Technical skills

Programming and tools: Python, Pandas, PyTorch, XGBoost, Git, OpenAI Gym; SQL (foundational knowledge).

Methods: Time-series and probabilistic forecasting, Transformers, graph learning, multi-task learning, reinforcement learning, clustering and benchmark evaluation.

Education

PhD researcher | TU Delft

June 2023–present | Expected completion Q2 2027

MSc Computational Science | University of Amsterdam

July 2021–June 2023

BSc Econometrics and Operations Research | Erasmus University Rotterdam

2018–2021

Leadership and collaboration

- Supervised three master's projects: unloading-time forecasting for logistics routing with Albert Heijn; RL traffic-signal control with Sweco and the City of Almelo; and availability-informed e-scooter pricing within the SUM project.
- Teach reinforcement learning in transportation and logistics to TU Delft master's students (2023–present). Co-established and co-maintain the SUM Lab GitHub presence for sharing research outputs.

Selected research outputs

Published as Jingyi Cheng. Full publication list

T-STAR — Probabilistic demand forecasting. Transportation Research Part C (2026). Article | Code

A Short-Term Predict-Then-Cluster Framework for Meal Delivery Services. Data Science for Transportation (2025). Article | Code

Real-Time Integrated Dispatching and Idle Fleet Steering with Deep Reinforcement Learning for a Meal Delivery Platform. Preprint (2025). Preprint

Languages

English — professional working proficiency | Chinese — native
Dutch — A2, actively improving